Quiet Neurons: a BDH quietens on what it just learned, not on what is merely predictable

DataForge 2026, Pathway track · concept **Sparse Non-Negative Activations** · Nilay Toshniwal, solo · artifact **nilaymastaadmi.github.io/quiet-neurons** · code **github.com/nilaymastaadmi/quiet-neurons**

THE DESIGN PRESSURE

A dense Transformer evaluates every feed-forward unit for every token, hard or not; activation sparsity is the counter-proposal. But Transformer sparsity is already emergent, arriving with training rather than by design [3]; it needs a hard rectifier, because only that yields exact zeros [4]; and *which* units fire is input-dependent, not merely how many [5]. Sparsity alone distinguishes nothing. What distinguishes an architecture is what its sparsity is a function of.

WHAT BDH CHANGES

Dragon Hatchling (BDH) [1] makes it structural. In the BDH-GPU formulation Pathway open-sources, every layer multiplies two rectified (ReLU) vectors elementwise, and a product of two sparse non-negative vectors is sparser than either. *Active* means a strictly non-zero entry in that product; a rectifier cannot emit a negative, so an active neuron is a positive amount of something, not a signed coordinate, which is what lets it be read as a concept. That product is the paper's $y_{t,l}$: Eq. (8) defines it as $(D_y LN(\rho x))_+ \odot x$, Appendix E's listing reads `relu(ln(a) @ decoder_y) * x`, and Fig. 14 counts its non-zeros. Mind the naming collision: the released code's `y_sparse` is the paper's rectified *intermediate*; the paper's y is the code's `xy_sparse`. Attention is the second change: query and key are the same tensor, so the score matrix is a causally-masked Gram matrix of activations with **no softmax**. Positions whose neurons fired alike bind to each other.

THE CLAIM, AND THE CORRECTION IT NEEDED

Section 6.4 reports that "fewer neurons are active ... for more predictable input signals": at layer 2 of a 65,536-neuron model, 4.0–7.5% active while memorising a new word against about 2.5% while repeating it, a 1.6–3.0× ratio [1, Fig. 14]. We reproduced that protocol on three models trained here from the public code, all for the same 2,309 steps, and it survives a 32-fold shrink: **1.47×, 2.12× and 1.87×** at 2,048, 8,192 and 16,384 neurons, each averaged over 1,280 sequences across five pinned samples, spread ±0.005 as a half-width (tables here quote it as a full min-to-max range). **In aggregate the paper is right: predictable text is quieter.** What follows sharpens that rather than contradicting it.

The same measurement narrows what that result can mean. Two blocks of one sequence are both predicted almost perfectly yet differ 2.65-fold in activity: the 13-letter warm-up runs at **9.7%** of layer 2 active at 0.0004 nats of mean surprise (a nat is the natural-log unit of cross-entropy; zero is perfect prediction), the 56 repeated letters at **3.6%** at 0.004 nats. The warm-up is identical in every training sequence, so it lives in the **weights**; the repeated word is new each time, so it can only come from the **context**. What separates them is not predictability but **where the knowledge came from**. We show activity tracks that; we do not show why. A learner falsifies it in a minute: inject an unpredictable letter mid-repetition and activity rises there, while the letter before is unchanged to nine decimals, because the model reads only leftwards.

System	Where its sparsity comes from	Carried per token	Evidence status
Dense Transformer feed-forward	Emergent in training [3]; needs a hard rectifier for exact zeros [4]	A key-value cache, growing with context	Extensive, independent
BDH-GPU, measured here	Structural: elementwise product of two rectified vectors, sparser than either	A fixed-size synaptic state	Reproduced here at toy scale; author-reported at 65,536 neurons
BDH-CQ	Not public	Not public	Developer-reported only

EVIDENCE, LABELLED

Reproduced here: the three ratios, the counterexample, the surprise-injection check, and parity of the in-browser model against PyTorch (logit max difference 6.1e-5, twelve of twelve sparsity series bit-identical). **Developer-reported, not reproduced:** the Fig. 14 band; 97.4% on Sudoku Extreme, which Pathway's own repository says comes from their internal implementation and does not reproduce from the open code; the 1B-to-600B scaling experiments; and BDH-CQ's 29.5% pass@2 on ARC-AGI-1 at $0.0007 per task [2]. **Not modelled:** BDH-CQ's internals, "proprietary" by its authors' own statement [2]; **it has no direct role here**, and inventing one would be worse than saying so. The strongest external grounding Pathway offers is development on Amazon SageMaker HyperPod, a commercial partnership rather than an independent evaluation. **Where it is likely worse:** sparse non-negative activations plausibly cost capacity against dense superposition, and no published comparison runs past GPT-2 scale.

LIMITATIONS, MATURITY, AND WHAT WOULD SETTLE THEM

Depth matters: layer 0 runs *backwards* at all three sizes (0.80, 0.80, 0.86), layer 3 is flat below 16,384 and weakly positive (1.11×) there. Scaling is **not monotonic**, and we published the opposite before the third model finished. The obvious confound is gone: the three had been cut by wall clock at 1,854, 1,917 and 2,309 steps of one schedule, so we retrained the short two to 2,309. Both rose, and the dip survived: the 8,192-to-16,384 gap of 0.246 is **40**× the mean half-width above. Still uncontrolled is that none finished the 4,000-step schedule; all stop at 58% of it. These are toys, 0.4M to 6.3M parameters on a synthetic task, as the paper's protocol is. Surprise and sparsity do *not* track per letter (8,192 neurons, layer 2: Pearson 0.30, Spearman −0.05), which killed a stronger claim. Most importantly we establish the *signature*, not its cause: nothing explains why context-held knowledge needs fewer neurons, and the control that would settle it relocates identical content between weights and context. **Maturity:** credible and outsider-checkable for this narrow property; the wider programme is earlier, its headline benchmarks developer-reported.

[1] Kosowski, Uznański, Chorowski, Stamirowska, Bartoszkiewicz, *The Dragon Hatchling*, arXiv:2509.26507 (2025), §6.4 and Fig. 14. [2] Engdahl, Kosowski, Chorowski et al., *BDH-CQ*, arXiv:2608.09888 (2026). [3] Li, You, Bhojanapalli et al., *The Lazy Neuron Phenomenon*, arXiv:2210.06313 (2022), ICLR 2023. [4] Mirzadeh, Alizadeh, Mehta et al., *ReLU Strikes Back*, arXiv:2310.04564 (2023). [5] Liu, Wang, Dao et al., *Deja Vu*, arXiv:2310.17157 (2023), ICML 2023.

Where to continue, in order. [1] §6.4 and Fig. 14 for the result reproduced here, then [1] §3.2 and Eq. (8) for the equations defining what is counted, then [3] for why sparsity alone distinguishes no architecture, then the artifact for the counterexample, which no paper states. — Reproduce every number: **python measure.py --ckpt checkpoints/bdh_n2048.pt --embd 64 --mult 32 --repeats 5**.